

Week 6 Final Baseline Model Report

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Project: Caribbean ED Validation of Machine-Learning Triage Support

Dataset: Mercer Emergency Department Triage Dataset

Random seed: 42

Objective and Data Preparation

The purpose of this work was to develop and assess two simple models for predicting Emergency Severity Index (ESI) triage level. A logistic regression model and a decision tree were compared with a basic model that guessed according to how often each triage level appeared in the dataset. This provided a fair starting point for determining whether the models performed better than chance before more advanced methods were explored.

The dataset contained 55,121 patient records. Ten pieces of information available during triage were selected, including age, oxygen-related measurements, vital signs, glucose and selected chief-complaint indicators. The models were trained to predict the recorded ESI level, ranging from Level 1 for patients requiring the most immediate care to Level 5 for the least urgent patients.

The data was divided into an 80% training set and a 20% testing set. The proportion of patients in each ESI level was kept similar in both sets, resulting in 44,096 records for training and 11,025 records for testing. A fixed random seed of 42 was used so that the same split and results could be reproduced. For logistic regression, the numerical measurements were placed on a comparable scale using information from the training data only. While, the decision tree used the original values since this type of model does not require scaling.

Baseline Models and Results.

Logistic regression was selected as the first baseline because it is a simple model whose results can be examined and explained. The patient measurements were placed on a comparable scale before training, and the model was allowed up to 1,000 iterations to complete the training process reliably. A decision tree was selected as the second baseline because it makes predictions through a sequence of decision rules. The tree was limited to five levels to keep it understandable and reduce the risk of learning patterns that may only apply to the training data.

Both models were assessed using the same unseen testing set. Their results were compared with a basic model that guessed according to how frequently each ESI level appeared in the dataset.

Table 1: Comparison of baseline model performance on the test dataset.

Model	Accuracy	Macro F1	Weighted F1	ESI Level 1 Recall
Stratified Random Baseline	0.375	0.204	0.375	0.000
Logistic Regression	0.545	0.221	0.468	0.000
Decision Tree	0.543	0.215	0.459	0.000

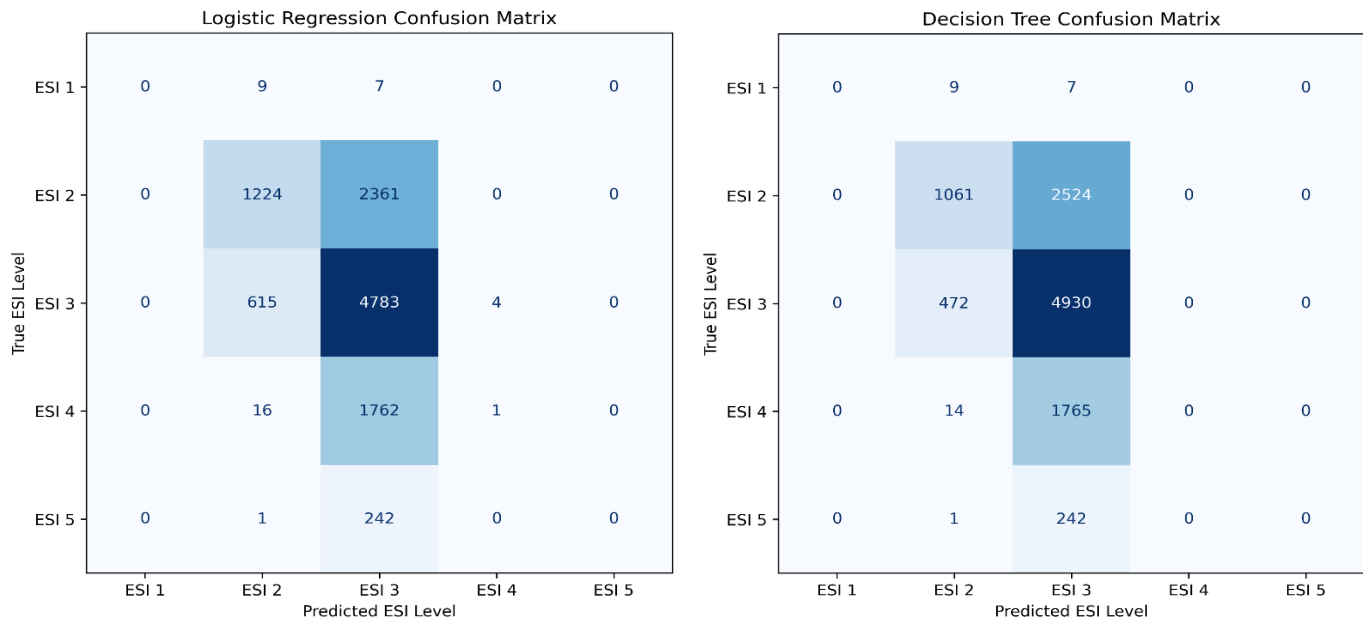


Figure 1: Confusion matrices for the baseline logistic-regression and decision-tree models on the test set.

Both models achieved higher overall accuracy, macro F1 and weighted F1 scores than the random-guess baseline. Logistic regression performed slightly better overall, with an accuracy of 54.5%, compared with 54.3% for the decision tree. However, this small advantage did not extend to the most critical patients. Neither model correctly identified any of the 16 ESI Level 1 patients in the testing set. Nine were classified as ESI Level 2 and seven as ESI Level 3. The models also relied heavily on the two largest groups, particularly ESI Level 3. This means that their improved overall scores mainly reflected better performance among the more common triage levels, rather than safe identification of the rarest and most urgent patients. Logistic regression was therefore the slightly stronger overall baseline, but neither model was clinically reliable for recognising ESI Level 1 patients.

Primary Metric and Clinical Relevance

ESI Level 1 recall was selected as the primary performance metric because it indicates the proportion of genuinely critical patients correctly identified by the model. In an emergency department setting, failing to recognise an ESI Level 1 patient presents a greater clinical risk than assigning a less urgent patient to a higher-priority category. A missed Level 1 case may result in delayed assessment, resuscitation or other immediate life-saving treatment.

Overall accuracy was not selected as the primary metric because the dataset was imbalanced, with most records belonging to ESI Levels 2 and 3. As a result, a model could achieve a reasonable overall accuracy by performing well on these more common classes while still failing to identify rare ESI Level 1 cases. This was observed in both baseline models, which achieved accuracies above 54% but did not correctly identify any of the 16 Level 1 patients in the testing set.

Macro F1 and weighted F1 were also considered to provide a broader assessment of model performance. Macro F1 assigns equal importance to each ESI level, making weak performance among rare classes more visible. Weighted F1 gives greater influence to classes with more patient records, meaning that stronger performance on ESI Levels 2 and 3 can increase the overall score. For example, logistic regression achieved a macro F1 of 0.221 and a weighted F1 of 0.468. This difference demonstrates why weighted performance should not be interpreted in isolation when the safe identification of uncommon but critically ill patients is the main clinical concern.

Failure Modes and Additional Refinement

The main failure was the inability of both baseline models to recognise the less common ESI categories, particularly ESI Level 1. Each model produced an ESI Level 1 recall of 0.000 and missed all 16 critically urgent patients in the testing set. Nine were assigned to ESI Level 2 and seven to ESI Level 3. This represents under-triage, as patients requiring immediate intervention were placed in lower-priority categories. The confusion matrices also showed that predictions were concentrated in the most common classes. Logistic regression predicted mainly ESI Levels 2 and 3, correctly identifying only one Level 4 patient and no Level 5 patients. The decision tree assigned every patient to either Level 2 or Level 3. This suggests that both models were strongly influenced by the unequal class distribution and were unable to distinguish the less common categories reliably. The small number of ESI Level 1 cases also limits the certainty of the findings. Only 16 Level 1 patients were included in the testing set, so each prediction had a large effect on recall and performance may vary with a different data split. The results should therefore be interpreted as an initial benchmark rather than a reliable estimate of clinical performance. An additional logistic-regression model using a broader set of triage-time information improved ESI Level 1 recall to 0.250 by correctly identifying 4 of the 16 critical patients. However, it still missed 12, including one patient assigned to ESI Level 4. Although this showed some improvement, the remaining missed cases mean that the model is not suitable for clinical use without further testing, imbalance handling and clinical review.

Conclusion

The Week 6 objective was achieved through the development and evaluation of the required logistic-regression and decision-tree models, together with comparison against a stratified random-guess baseline. Logistic regression performed slightly better overall, but neither baseline model identified any ESI Level 1 patients. These findings demonstrate why overall accuracy alone is not sufficient for evaluating safety in an imbalanced triage dataset. Although both models achieved accuracies above 54%, this was mainly due to stronger performance on the more common ESI Levels 2 and 3. Their failure to recognise the most urgent patients means that they cannot be considered clinically reliable. The refined logistic-regression model had increased ESI Level 1 recall but missed 12 of the 16 critically urgent patients. The models should therefore be treated as experimental benchmarks rather than patient-care tools. Future work should explore class weighting, careful resampling, further development of triage-time predictors and repeated testing across different data splits. ESI Level 1 recall should remain the main safety measure, supported by macro F1 and confusion matrices. Any future model would also require emergency-clinician review and validation using local Mercer General Hospital data before workflow testing.